d if we fix α (usually $\alpha \in [0.5, 2.5]$). We write the PHD calculation as $d = \text{PHD}(W)$ to simplify our notation.

Problems in Estimating PHD for Short Texts (Local Density Peak). Although estimating PHD with increased sampling is sample-efficient, it introduces additional challenges for short texts, as their point cloud contains only a small number of points. When the number of points is small, there is a higher probability that these points are too close (or distinct) to each other. This could cause $\log E_\alpha^0(W_i)$ to become unstable due to the edge lengths of the MST becoming extremely small (or large). We refer to this issue as a local density peak (Tulchinskii et al., 2024). Meanwhile, as previously discussed, the value of PHD is based on the slope of the linear regression between $\log n_i$ and $\log E_\alpha^0(W_i)$. Therefore, an unstable $\log E_\alpha^0(W_i)$ caused by local density peaks could affect the accuracy of the PHD estimation process. An example is shown in Figure 1. Specifically, the regression contains three sampled subsets through increased sampling, which contain 10, 20, and 40 points (before logarithm), where $\log 20$ experiences a local density peak. The peak will lead to a higher slope and thereby lead to a higher PHD value d. In practice, the increased sampling process is conducted multiple times to reduce the influence of local density peaks (Birdal et al., 2021). However, if we only have a few points to sample, the local density peak cannot be easily avoided by the randomness in sampling.

3 Method: Short-PHD

In this section, we first introduce a technique named off-topic insertion (OCI) which can stabilize the PHD estimation for short texts. We then present Short-PHD, an OCI-based method for detecting short LLM-generated texts. Finally, we provide empirical evidence to validate the effectiveness of these two techniques.

3.1 OCI: Off-topic Content Insertion

As discussed in Section 2.2, compared to long texts, the PHD method is more likely to encounter local density peaks when estimating short texts, leading to unstable results that may be either too high or too low. To address such an issue, we propose a hypothesis using off-topic insertion as follows:

Hypothesis 1 (OCI). *Off-topic content insertion (OCI) will stabilize and increase the PHD values of short texts.*

The rationale behind this hypothesis is twofold. First, by adding additional content, the text becomes longer, helping the PHD estimation avoid local density peaks and achieve more stable results. Second, since PHD measures the connectedness of the embedding, adding completely off-topic text reduces this connectedness, thereby increasing the PHD for any given input text. For example, if the original text is a Wikipedia page on the American Civil War, the off-topic content could be a tip for detecting LLM-generated texts. In practice, we insert a fixed text sequence before each text sample. We apply OCI in distinguishing short human-written and LLM-generated texts below.

3.2 Short-PHD: Detecting Short LLM-generated Text

We assume that the PHD distributions of short human-written and LLM-generated texts follow independent normal distributions, i.e., $d^h \sim N(\mu^h, (\sigma^h)^2)$ and $d^m \sim N(\mu^m, (\sigma^m)^2)$, where the superscripts h and m signify human and machine, respectively. Following Tulchinskii et al. (2024), human-written texts exhibit larger PHDs than LLM-generated texts. Therefore, we can use PHD to detect LLM-generated text by measuring the probability that a human-written text has a higher PHD than an LLM-generated text, i.e., $P(d^h > d^m)$. Specifically, let $Z = d^h - d^m$. Then, the probability of d^h being greater than d^m is given by $P(d^h > d^m) = P(Z > 0)$. Based on our assumption, d^h and d^m are independent, then Z follows a normal distribution $Z \sim N(\mu^h - \mu^m, (\sigma^h)^2 + (\sigma^m)^2)$. Thus, the standardized Z follows a standard normal distribution. The probability of correctly classifying a text is

Algorithm 1 Short-PHD

Input: a text sequence X, an off-topic content set T_X, an pretrained LLM P_M, a PHD estimator PHD(), and a detection threshold ϵ.
Initialize a score list S
for each off-topic content $x_t \in T_X$ **do**
  $X_t \leftarrow \text{concat}(x_t, X)$
  Compute the embedding of X_t, $W_t \leftarrow P_M(X_t)$
  Estimate the PHD of W_t, $d_t \leftarrow \text{PHD}(W_t)$
  Record d_t in S
end for
Score $s \leftarrow \text{mean}(S)$
if $s > \epsilon$ **then**
  return `human-written`
else
  return `LLM-generated`
end if

given by

$$P(Z > 0) = \Phi\left((\mu^h - \mu^m)/\sqrt{(\sigma^h)^2 + (\sigma^m)^2}\right), \tag{2}$$

where Φ is the cumulative distribution function (CDF) of a standard normal distribution. Since Φ is monotonically increasing, a larger mean difference $(\mu^h - \mu^m)$ and lower variance $((\sigma^h)^2 + (\sigma^m)^2)$ increase the probability of successfully detecting LLM-generated text.

When the text is short, the estimation of PHD becomes unstable, and the variance $((\sigma^h)^2 + (\sigma^m)^2)$ increases, making detection ineffective. Therefore, in Short-PHD, we apply OCI to stabilize the estimation process for detection. Similarly to the previous PHD method analysis, we assume that each PHD distribution after OCI follows independent normal distributions: $d_I^h \sim N(\mu_I^h, (\sigma_I^h)^2)$ and $d_I^m \sim N(\mu_I^m, (\sigma_I^m)^2)$, where I denotes OCI. According to Hypothesis 1 on OCI, the variance of PHD decreases ($\sigma_I^h < \sigma^h$ and $\sigma_I^m < \sigma^m$), while the PHD values for both human-written and LLM-generated texts increase. We approximate this increase as follows $\mu_I^h \approx \mu^h + \epsilon^h$ and $\mu_I^m \approx \mu^m + \epsilon^m$, where ϵ^h and ϵ^m represent the increases in PHD for human-written and LLM-generated texts, respectively. In Short-PHD, we further make the following assumption on the increase.

Hypothesis 2 (Short-PHD). *The gap between human-written and LLM-generated PHD values remains the same in expectation after inserting the same off-topic content.*

This implies that $\mu^h - \mu^m \approx \mu_I^h - \mu_I^m$, which means the PHD scores of human-written and LLM-generated texts increase by the same value $\epsilon^h \approx \epsilon^m$. Therefore, in Short-PHD, when the gap between human-written and LLM-generated texts remains and the variances are decreased, we have $P(d^h > d^m) < P(d_I^h > d_I^m)$, meaning that LLM-generated texts can be detected with a higher probability using Short-PHD compared to PHD. We present Short-PHD in Figure 1 and Algorithm 1. In practice, Short-PHD employs OCI multiple times with different content and computes the average.

3.3 Empirical Validation

We conduct an empirical validation to test the proposed hypotheses on OCI and Short-PHD using the same dataset from the original PHD paper (Tulchinskii et al., 2024). Specifically, we choose two subsets from Tulchinskii et al. (2024): Reddit WritingPrompts (WP) stories (Fan et al., 2018) and Wikipedia pages (Guo et al., 2020), both of which contain GPT-3.5-generated and human-written texts. We leverage LLaMA-3-8B (Dubey et al., 2024) to compute the embedding $W = \text{LLaMA}(X)$ of a text sequence X and calculate the PHD detection score using increased sampling. We then apply OCI before the input text and calculate the PHD score for the modified text. Specifically, we choose off-topic content related to tips for

Table 2: Average number of tokens in each dataset and statistics of detection score distributions in each dataset with PHD and Short-PHD. Bolded standard deviation (std) implies large std reduction by Short-PHD.

Dataset	WP human	WP GPT-3.5	Wikipedia human	Wikipedia GPT-3.5
Token number	153	276	461	165
PHD mean	10.17	8.84	10.70	8.93
PHD std	1.49	0.79	1.01	1.54
Short-PHD mean	10.26	9.23	10.81	9.23
Short-PHD std	**0.79**	0.67	0.85	**1.00**

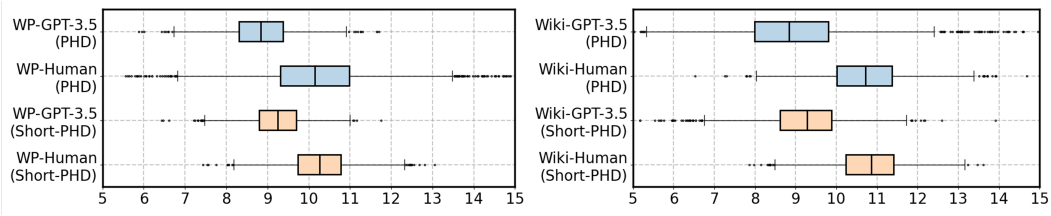

Figure 2: Boxplots of score distributions on human-written and GPT-3.5-generated Wikipedia and WritingPrompts (WP) data.

detecting LLM-generated texts, which is generally unrelated to the text being detected. The off-topic content is shown in Figure 1. We provide the average number of tokens in each subset and the statistics of PHD scores in Table 2. Additionally, we present the distributions of PHD scores using boxplots in Figure 2.

Validation of Hypothesis 1 (OCI). From Table 2, we first observe that the average token number of human-written WP and GPT-3.5-generated Wikipedia is significantly smaller than that of human-written Wikipedia and GPT-3.5-generated WP. Meanwhile, we also find that subsets with shorter texts have a larger standard deviation than those with longer texts. Similarly, as depicted in Figure 2, it is demonstrated that human-written WP and GPT-3.5-generated Wikipedia have wider (blue) boxes compared to other blue boxes, indicating larger variances. Therefore, we can validate that the PHD estimations are less stable for both short human-written and short LLM-generated texts than for long texts.

Next, as shown in Table 2, compared to the standard deviation of PHD, applying OCI significantly reduces the standard deviation of PHD values for both short human-written WP and GPT-3.5-generated Wikipedia. The same results can be concluded from Figure 2, where the yellow boxes (Short-PHD) of human-written WP and GPT-3.5-generated Wikipedia are narrower than the blue boxes of these two subsets. In summary, we conclude that OCI stabilizes the estimation of PHD for short texts. Second, we observe that the mean PHD values for all subsets simultaneously increase. Therefore, we also conclude that applying OCI increases the PHD values for both short human-written and LLM-generated texts.

Validation of Hypothesis 2 (Short-PHD). As depicted in Figure 2, the gap between human-written and LLM-generated text after applying OCI is largely maintained compared to the PHD method, without dropping significantly. For example, in the WP dataset, the gap for PHD is 1.33 $(10.17 - 8.84)$, while the gap for Short-PHD is 1.03 $(10.26 - 9.23)$. Similarly, in the Wikipedia dataset, the gap for PHD is 1.77 $(10.70 - 8.93)$, whereas for Short-PHD, it is 1.58 $(10.81 - 9.23)$. Therefore, the Short-PHD method could establish a detection threshold like the PHD method to identify LLM-generated texts. Furthermore, by plugging in the mean and standard deviation values, we can explicitly compute the value of the CDF. For the PHD method, the probabilities of successful detection are approximately $\Phi(\frac{10.17-8.84}{\sqrt{1.49^2+0.79^2}}) \approx \Phi(0.79) \approx 0.78$ and $\Phi(0.96) \approx 0.83$ for the WP and Wikipedia datasets, respectively. After applying Short-PHD, the probabilities increase to $\Phi(0.98) \approx 0.84$ and $\Phi(1.20) \approx 0.88$ for the WP and Wikipedia datasets, respectively. This indicates that the